

Aditya S

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Thiruvananthapuram, Kerala, India | Open to remote

SUMMARY

Product engineer at UST, building web products backed by AWS Bedrock. Founder and lead maintainer of open-source developer tools used by over 1 million people across 25,000+ deployments, and a member of Anthropic's Claude for Open Source Program. Works across the whole stack: Angular front end, Spring Boot APIs, and the retrieval and guardrail layer behind the model.

EXPERIENCE

Associate 2, Product Engineering

UST

Jul 2025 to Present

Thiruvananthapuram, India

- Promoted from Associate 1 to Associate 2 in August 2026.
- Build and maintain sales and operations features on an enterprise telecom platform, working across the Angular front end and the Spring Boot REST services behind it.
- Led the front-end migration from **Angular 15 to Angular 21**, rewriting a large share of the components and shared modules to fit standalone components, the new control flow, and signals.
- Deliver generative-AI features on AWS: retrieval pipelines on **Amazon Bedrock Managed Knowledge Bases**, policy enforcement through **Bedrock Guardrails**, and product features backed by Claude Sonnet.
- Cut API response times by up to **40%** by restructuring REST endpoints, tightening DTO boundaries, and rewriting inefficient database queries.
- Produce product demo films in Remotion (React), scripted and rendered from code, for expo booths and client briefings.

Tech Intern

BreadcrumbsAI

2024

Remote

- Built Python scraping services with Playwright for dynamic extraction and BeautifulSoup for parsing. Added structured logging so long-running jobs failed loudly instead of dropping records silently.

Project Lead & Backend Developer

Google Developer Student Clubs, MBCET

2022 to 2024

Kerala, India

- Led club technical projects; built the GDSC MBCET website backend and event automation tooling.

Campus Lead

GTECH μ Learn, MBCET

2023 to 2024

Kerala, India

- Ran campus learning and skill-development programmes; campus crossed 1 million karma points.

OPEN SOURCE

Ultroid: Telegram Userbot Framework

Founder & Lead Maintainer, 2021 to Present

- Founded and lead a Python automation framework with **3k+ GitHub stars** and **25,000+ deployed instances** worldwide.
- Designed the plugin architecture that carries 200+ modules, so contributors can extend the runtime without touching core.
- Engineered a broadcast system reaching **~1.6M users**: async Python with **motor**, token-bucket rate limiting to stay inside API quotas, SQLite for resumable delivery state, and **_id**-based cursor pagination over MongoDB Atlas.
- Run releases, code review, and issue triage for a distributed contributor base. github.com/TeamUltroid/Ultroid

Stack: Python, MongoDB, SQLite, Docker, Shell

ChannelActions: Telegram Bot

Creator & Maintainer

- Automates Telegram channel join requests (auto-approval, auto-decline, custom welcome flows) for **1M+ active users**, localised into 25+ languages.
- Rewrote the service on Deno with Deno KV, and shipped a web dashboard so admins configure it themselves.
- channelactions.xditya.me | github.com/xditya/ChannelActionsBot

Stack: TypeScript, Deno, Deno KV, MongoDB, i18n

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, Java, Kotlin, Dart, C++, SQL
Frontend	Angular, React, Next.js, Vite, Remotion
Mobile	React Native, Expo, Flutter, Jetpack Compose, Swift and Kotlin native modules
Backend & Data	Spring Boot, FastAPI, Node.js, Deno, REST API design, MongoDB Atlas, SQLite, Deno KV, Vercel KV
Cloud & AI	AWS, Amazon Bedrock (Knowledge Bases, Guardrails, Claude), RAG pipelines, MCP, Gemini API, Agno
Tooling	Git, GitLab CI/CD, GitHub Actions, Docker, ELK, Vercel, ffmpeg, L ^A T _E X

SELECTED PROJECTS

engram: Local-First Knowledge Library

Android & iOS, AGPL-3.0 | Aug 2026 to Present

- Open-source app for saving links, articles, images and notes, with no account and no server. Every device holds a full SQLite copy, and everything exports in one tap to files, JSON, CSV, or Obsidian markdown.
- Built **end-to-end encrypted sync** that runs through the user's own Google Drive, iCloud or WebDAV: an append-only op-log with per-field last-writer-wins over hybrid logical clocks, content-addressed blobs, XChaCha20-Poly1305, and BIP39 recovery phrases. Property and fuzz tested.
- One-tap capture from any app, using a translucent overlay activity on Android and an iOS share extension that boots a slim React Native root inside the extension process.
- Tagging and search work with no model at all. Optional intelligence layer adds summaries and visual search through a bring-your-own API key or fully on-device inference (ExecuTorch, Qwen3 0.6B with MiniLM embeddings).
- Full-text search over saved page content via SQLite FTS5, with **tag:**, **site:** and **before:** operators; saved searches become Spaces.
- engram.xditya.me | github.com/xditya/engram

Stack: TypeScript monorepo (pnpm), React Native, Expo SDK 57, op-sqlite, Swift and Kotlin native modules, GitHub Actions CI

React, TypeScript, Vite

Splitty: Bill-Splitting PWA

React, TypeScript, Vite

- Splits restaurant bills from a photo. OCR falls back through three tiers: a user-supplied Gemini key, then a shared server-side proxy, then **tesseract.js** as WASM in the browser, so the app degrades instead of failing.
- Wrote an integer-paise allocator using the largest-remainder method, fuzz-tested so splits reconcile exactly to the bill total with no rounding drift.
- Handles GST, liquor VAT at its own rate, and service charge applied before tax. Settle-up runs through UPI QR codes, and sessions are shareable via Vercel KV. splitty.xditya.me | github.com/xditya/Splitty

Campus Services Management System

Final Year Project

- Print automation platform where students upload documents, watch the print queue update live, and pay from an integrated wallet. A digital approval workflow handles campus forms and notifies faculty.
- Integrated a QR-based vending machine on ESP32 hardware for food distribution, delivered inside a cross-platform Flutter app.
- github.com/xditya/CampusServicesManagementSystem

Stack: Dart, Flutter, TypeScript, Deno, MongoDB, C, ESP32, ESP32-CAM

Telethon Snippets: VS Code Extension

5k+ installs

- Snippet pack for the Telethon library, published on the Visual Studio Marketplace and installed over 5,000 times. marketplace.visualstudio.com/items?itemName=xditya.telethon

EDUCATION

B.Tech, Electronics and Communication Engineering

2021 to 2025

Mar Baselios College of Engineering and Technology

Kerala, India

CERTIFICATIONS & ACHIEVEMENTS

- **AWS Certifications:** Amazon Bedrock Managed Knowledge Bases; Amazon Bedrock Guardrails.
- **Anthropic Claude for Open Source Program**, selected member.
- **Winner, EDEN 3.0 Hackathon** (Marian Engineering College, 2024) for an adaptive employee-management platform with tracking and automated promotion calculation.
- **Top 100 Coders, Kerala** (μ Learn at HuddleGlobal, 2023), a state-level initiative recognised by the Government of Kerala.
- **#12 in India** for public GitHub contributions, 2021; 660+ followers on GitHub.